Exercise 1: Disease Spread (Best for ABM)

Scenario

Simulate how a virus spreads among people.

Agents

- Green = Healthy
- Red = Infected

Code

```
turtles-own [infected?]

to setup
  clear-all

  create-turtles 50 [
    setxy random-xcor random-ycor
    set infected? false
    set color green
  ]

  ask one-of turtles [
    set infected? true
    set color red
  ]

  reset-ticks
end

to go

  ask turtles [
    rt random 30
    lt random 30
    fd 1
  ]

  ask turtles with [infected?] [

    ask other turtles in-radius 1 [

      if not infected? [
        set infected? true
        set color red
      ]

    ]
  ]

  tick
end
```

Learning

- Interaction between agents
- Epidemic modeling
- ABM concepts

Exercise 2: Traffic Simulation

Scenario

Cars move on a road and avoid collisions.

```
to setup

  clear-all

  create-turtles 20 [
    setxy random-xcor 0
    set heading 90
  ]

  reset-ticks
end

to go

  ask turtles [

    if not any? other turtles-on patch-ahead 1 [
      fd 1
    ]

  ]

  tick
end
```

Learning

- Traffic flow
- Collision avoidance

Exercise 3: Queue at a Bank

Scenario

Customers wait for service.

```
to setup
```

```
  clear-all

  create-turtles 20 [
    setxy random-xcor random-ycor
  ]

  reset-ticks
end

to go

  ask turtles [

    face patch 0 0

    if distancexy 0 0 > 1 [
      fd 0.5
    ]

  ]

  tick
end
```

Learning

- Queue behavior
- Crowd movement

Exercise 4: Ant Food Collection

Scenario

Ants search for food and bring it home.

Simple Rules

- Search randomly
- Find food
- Return to nest

This is a famous NetLogo example.

Learning

- Swarm intelligence
- Collective behavior

Exercise 5: Airport Passenger Movement

This matches one of your lecture papers.

Scenario

Passengers:

- Enter airport
- Visit shops
- Move toward gate

Rules

```
Passenger sees shop
↓
Visit shop
↓
Spend time
↓
Move to boarding gate
```

Learning

- Human behavior modeling
- Decision making

Exercise 6: Museum Visitor Simulation

Scenario

Visitors:

- Enter museum
- Visit paintings
- Avoid revisiting exhibits
- Exit museum

Learning

- Memory
- Vision
- Navigation

Lab Exercise: Airport Passenger Shopping Behavior Model

Objective

To simulate passenger movement inside an airport terminal where passengers may visit shops before proceeding to their boarding gate.

Problem Statement

Passengers enter the airport.

Each passenger:

1. Has a boarding gate.
2. Can see nearby shops.
3. Decides whether to visit a shop.
4. Avoids overcrowded shops.
5. Eventually moves toward the boarding gate.

This demonstrates **Agent-Based Modeling (ABM)** and decision-making behavior.

Environment

Blue Patch

Airport entrance

Yellow Patches

Shops

Red Patch

Boarding gate

Turtles

Passengers

Setup Code

```
globals [gate]

turtles-own [
  shopping?
]

to setup

  clear-all

  ;; Gate
  ask patch 15 0 [
    set pcolor red
  ]

  ;; Shop 1
  ask patches with [
    pxcor > -5 and pxcor < 0 and
    pycor > 5 and pycor < 10
  ]
  [
    set pcolor yellow
  ]

  ;; Shop 2
  ask patches with [
    pxcor > -5 and pxcor < 0 and
    pycor < -5 and pycor > -10
  ]
  [
    set pcolor yellow
  ]

  create-turtles 30 [

    setxy -15 random-ycor

    set color blue

    set shopping? false
  ]

  reset-ticks
end
```

Passenger Behavior

```
to go

  ask turtles [

    ;; chance to shop

    if not shopping? [
```

```
      if random 100 < 20 [

        let target one-of patches with [pcolor = yellow]

        if target != nobody [

          face target

          set shopping? true
        ]
      ]
    ]

    ;; move toward gate

    if not shopping? [
      face patch 15 0
    ]

    fd 0.5

    ;; reached shop

    if shopping? [

      if [pcolor] of patch-here = yellow [

        wait 0.2

        set shopping? false
      ]
    ]
  ]

  tick
end
```

Expected Output

Passengers:

```
Entrance
    ↓

Passenger enters
    ↓

May visit shop
    ↓

Shop for some time
    ↓

Move toward gate
    ↓
```

```
Exit
```

You will observe:

- Some passengers go directly to the gate.
- Some stop at shops.
- Different passenger decisions emerge naturally.

Concepts Learned

Agent

Passenger

Environment

Airport terminal

Decision Rule

```
20% probability of shopping
```

Goal-Oriented Movement

```
face patch 15 0
```

Human Behavior Modeling

Passengers make individual decisions rather than following a fixed path.

Experiment: Heuristic-Based Airport Passenger Shopping Model

Objective

To develop an Agent-Based Model (ABM) in NetLogo that simulates passenger movement and shopping behavior inside an airport terminal.

Problem Statement

Passengers enter an airport terminal and move toward their boarding gate. Some passengers may decide to visit airport shops before proceeding to the gate. The decision to shop depends on the passenger's available budget.

Design a NetLogo model that simulates this behavior and analyze how passenger spending power affects shopping activity.

Tasks

Task 1: Create the Airport Environment

- Create an airport entrance.
- Create one boarding gate.
- Create at least two shopping areas using colored patches.
- Use different colors for gate and shops.

Task 2: Create Passenger Agents

- Generate 30 passenger agents.
- Place passengers at the airport entrance.
- Assign each passenger:
  - A random amount of money.
  - A shopping status variable.

Task 3: Implement Passenger Movement

- Passengers should move toward the boarding gate.
- Use NetLogo commands such as:
  - `face`
  - `fd`
  - `patch`

Task 4: Implement Shopping Decision

- Passengers with sufficient money should have a chance to visit a shop.
- Passengers with low budgets should proceed directly to the gate.
- Introduce a probability-based shopping decision.

Task 5: Implement Shopping Behavior

- When a passenger reaches a shop:
    - Reduce the passenger's money.
    - Record that shopping has occurred.
    - Resume movement toward the boarding gate.

Task 6: Collect Statistics

Display:

- Total number of passengers.
- Number of passengers who visited shops.
- Number of passengers who went directly to the gate.
- Average remaining money.

Advanced Tasks (Bonus)

Task 7: Passenger Categories

Create two passenger types:

- Shopper
- Non-Shopper

Compare their behavior.

Task 8: Crowded Shop Avoidance

If too many passengers are already in a shop:

- Passengers should select another shop.
- Or proceed directly to the gate.

Task 9: Time Constraint

Add a flight departure timer.

- If boarding time is near,
- Passengers should stop shopping and move directly to the gate.

Questions

1. What is an agent in this simulation?
2. What agent attributes are used in the model?
3. How does money influence passenger behavior?
4. Why is this model considered Agent-Based Modeling?
5. What happens if the shopping probability is increased?
6. How would adding more shops affect passenger flow?
7. How can this model help airport management?
8. What is the difference between a shopper and a non-shopper passenger?